Refraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a change in wave speed to wavelength and direction.

SUCCESS CRITERIA - I can...

1. Measure ray angles from the normal.
2. Explain frequency continuity at a stationary boundary.
3. Predict bending toward or away from the normal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “refraction” and “normal line” to explain your model or result. Sentence starters are optional.

refraction: change in wave direction when speed changes at a boundary

normal line: a reference line perpendicular to a surface

medium: the material through which a mechanical wave travels

01 Bring back an earlier idea

A wave snapshot has crests at 0.1, 0.4, 0.7 m.
Find wavelength. Its period is 0.05 s; find frequency.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $n_1 \sin\theta_1 = n_2 \sin\theta_2$; $n = c/v$; $v = f\lambda$.
Angles from the normal. Stationary boundary: frequency stays constant.

Worked reasoning: Light enters a slower medium at an angle to the normal and bends toward the normal. The normal is perpendicular to the boundary.

Your turn: Draw and label the boundary, normal, incident ray and refracted ray.

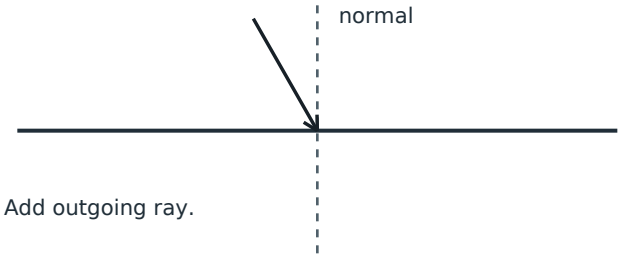
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Mark the normal at the boundary. A ray enters a slower medium at an angle.
Sketch the refracted direction.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

04 Solve and justify

Light travels from air (n=1.00) into glass (n=1.50) at 30° from normal.
Use $n_1\sin\theta_1 = n_2\sin\theta_2$ to find refracted angle.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student measures an incidence angle from the surface instead of the normal. If surface angle is 20°, what angle belongs in Snell’slaw?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

06 Choose a method

A 0.4 kg metal cools from 92 to 32°C while 0.3 kg water warms from 12 to 32°C.
Use water c=4200; neglect cup/losses.
Find metal c.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

07 Exit check

From glass (n=1.5) to air, find critical angle using $\sin\theta_c = 1/1.5$.
State what happens above that angle in ideal model.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

Light passes from n=1.0 to n=1.5 at 30° to the normal.
Use $n_1\sin\theta_1 = n_2\sin\theta_2$ to find refracted angle. Then find the critical angle for the reverse direction.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③